DT 111 · TOPIC 7

Ethics and Societal Impact of AI

Privacy, bias, fairness, transparency, safety, governance and accountability

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain why ethical considerations begin before modelling.
- Identify privacy and common sources of bias.
- Discuss fairness beyond overall accuracy.
- Explain transparency, explainability and accountability.
- Recognise robustness, security, monitoring and governance as lifecycle responsibilities.



Responsible AI begins with the problem

A technically accurate system can still be harmful if the goal, data, deployment context or consequences are poorly considered.

Responsible AI across the lifecycle

1. **Purpose** — Is the problem appropriate for AI?
2. **Data** — Permission, quality, representation
3. **Design** — Metrics, safeguards, human roles
4. **Test** — Performance and group impacts
5. **Deploy** — Controls and communication
6. **Monitor** — Drift, incidents and appeals

Privacy principles

Data minimisation

Collect only data reasonably necessary for the stated purpose.

Purpose limitation

Use data for clear legitimate purposes.

Security

Protect personal information from unauthorised access or misuse.

Retention

Do not keep personal data indefinitely without reason.

Where bias can enter

Historical bias

Past data may reflect previous inequalities or decisions.

Sampling bias

Collected data may not represent the intended population.

Measurement bias

Variables may measure some people less accurately than others.

Label bias

Targets may contain subjective or inconsistent judgements.

Case study: scholarship screening

Applicant	Prior school	Income data	Model decision
A	Urban	Complete	Shortlist
B	Rural	Missing	Reject
C	Urban	Complete	Shortlist
D	Rural	Partial	Reject

A pattern of rejections among rural applicants may reflect data availability rather than true suitability.

The model output should trigger investigation, not an automatic claim of fairness.

Fairness needs more than overall accuracy

Overall performance

- A model can be 90% accurate overall
- The same error rate may not apply to every group
- Aggregate metrics can hide unequal harms

Fairness questions

- Who receives benefits and harms?
- Are error rates very different across groups?
- Can people challenge an important decision?

Transparency and explainability

Transparency

Provide appropriate information about how a system is developed, used and limited.

Explainability

Provide understandable reasons or important factors behind an output where appropriate.

Communication

Users should know when AI is involved and what its limits are.

Human oversight and accountability

Oversight

Humans can review, intervene or override important automated outcomes.

Appeal

Affected people should have a route to challenge significant decisions.

Accountability

An organisation or person remains responsible for design, use and consequences.

Robustness, safety and security

Robustness

System should remain reliable under reasonable changes or noisy inputs.

Safety

Prevent or reduce foreseeable harm from failures.

Security

Protect against attacks, unauthorised access and manipulation.

Risk prioritisation

Risk management considers both how likely a problem is and how severe the consequences could be.

Likelihood ↓ / Impact →	Low	Medium	High
Low	Lower priority	Lower priority	Review / mitigate
Medium	Lower priority	Review / mitigate	High priority
High	Review / mitigate	High priority	High priority

Low-risk and high-risk use

Lower-stakes example

- Recommend learning resources
- Easy to ignore or correct
- Limited consequence of one error

Higher-stakes example

- Credit, hiring or medical triage
- Errors can affect rights or welfare
- Requires stronger testing, oversight and appeal

Governance makes responsibility operational

Policies, roles, documentation, testing requirements and monitoring turn ethical principles into concrete organisational practice.

Governance components

Policy

Rules for acceptable AI use and prohibited use.

Roles

Clear responsibility for data, models, deployment and incident response.

Documentation

Record datasets, assumptions, limitations and changes.

Monitoring

Track performance, drift, complaints and incidents after deployment.

Post-deployment monitoring

Data drift

The population or input data may change over time.

Performance drift

Accuracy or reliability may decline.

Unexpected use

Users may apply the system in ways not anticipated.

Incident learning

Failures should lead to correction and process improvement.

Core ethical questions for AI projects

- What problem is the system solving, and should AI be used?
- Whose data are collected, and with what permission?
- Who may be underrepresented or disadvantaged?
- What happens when the system is wrong?
- Who can review or appeal an important decision?
- Who is accountable after deployment?

Topic 7 key ideas

Ethics is part of system design, not an activity added after deployment.

Privacy, bias and fairness depend on the data and the social context.

Overall accuracy can hide unequal error patterns.

Human oversight, appeal and accountability are essential for important decisions.

Responsible AI includes robustness, safety, security, governance and ongoing monitoring.